

Design of Test Cases and a Maintenance Plan for a Sample Software Application

Aim

To prepare a software project plan for a sample application by defining the project scope, timeline, team roles, and project activities for successful software development.

Objective

- Understand software testing concepts.
- Design effective test cases.
- Verify software functionality.
- Prepare a software maintenance plan.
- Improve software reliability and performance.

Selected Software Application

Library Management System:

Overview

A Library Management System is developed to manage books, users, borrowing, returns, fines, and reports. Proper testing ensures that all modules work correctly before deployment, while maintenance keeps the software updated after release.

Test Cases

Test Case ID	Test Scenario	Expected Result
TC-01	User Login	Login Successful
TC-02	Invalid Password	Error Message Displayed
TC-03	Search Book	Book Details Displayed
TC-04	Borrow Book	Book Issued Successfully
TC-05	Return Book	Book Returned Successfully
TC-06	Fine Calculation	Correct Fine Displayed
TC-07	Add New Book	Book Added Successfully
TC-08	Delete Book	Book Removed Successfully

Testing Types

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing (UAT)
- Regression Testing

Maintenance Plan

Corrective Maintenance

- Fix software bugs.
- Resolve system errors.

Adaptive Maintenance

- Update software for new operating systems.
- Support new hardware.

Perfective Maintenance

- Improve software performance.
- Enhance user interface.
- Add new features.

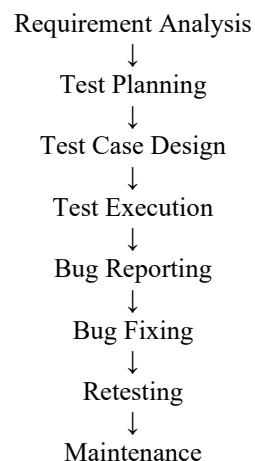
Preventive Maintenance

- Database Backup
- Security Updates
- Performance Monitoring
- Code Optimization

Maintenance Schedule

Activity	Frequency
Database Backup	Daily
Security Update	Monthly
Bug Fixes	As Required
Performance Monitoring	Weekly
Feature Enhancement	Quarterly

Test Execution Flow



Conclusion

Designing proper test cases and implementing a structured maintenance plan are essential for delivering high-quality software. Through systematic testing and regular maintenance, the Library Management System can provide reliable performance, improved security, and better user satisfaction throughout its lifecycle